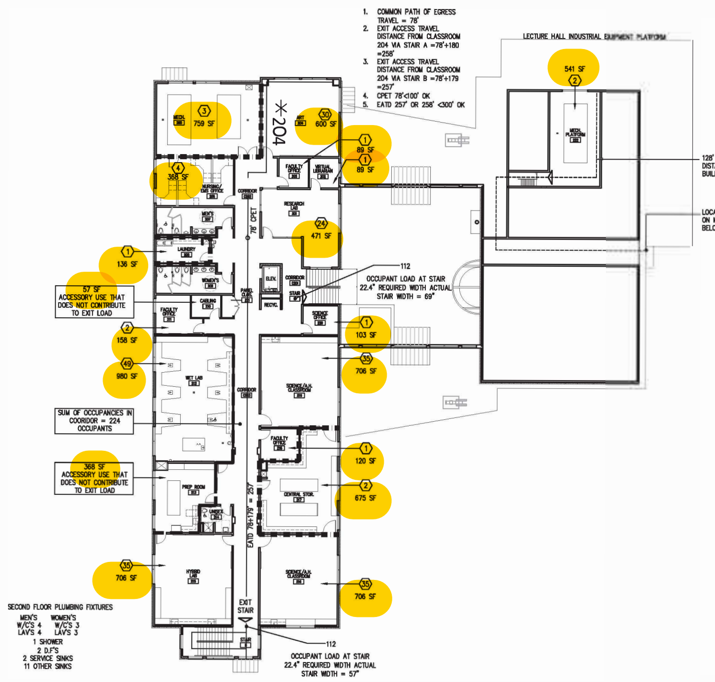
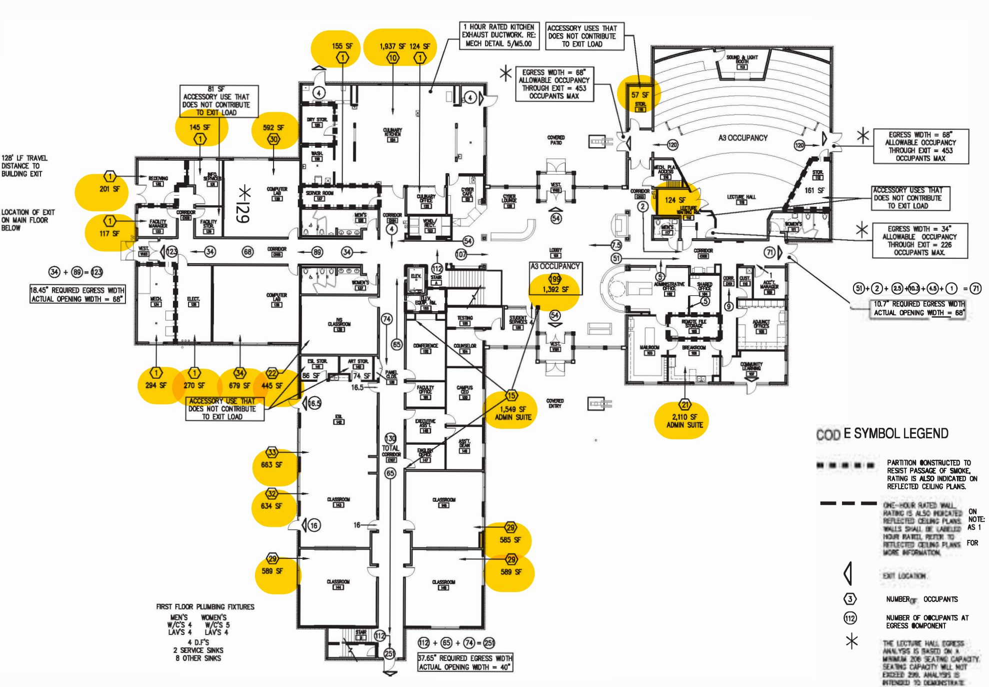

LECTURE HALL INDUSTRIAL EQUIPMENT PLATFORM
541 SF
1. COMMON PATH OF EGRESS TRAVEL = 78'
2. EXIT ACCESS TRAVEL DISTANCE FROM CLASSROOM 204 VIA STAIR A =78'+180 =258'
3. EXIT ACCESS TRAVEL DISTANCE FROM CLASSROOM 204 VIA STAIR B =78'+179 =257'
4. CPET 78'<100' OK
5. EATD 257' OR 258' <300' OK
128' LF TRAVEL DISTANCE TO BUILDING EXIT
LOCATION OF EXIT ON MAIN FLOOR BELOW
✳204
759 SF
600 SF
89 SF
89 SF
368 SF
471 SF
136 SF
57 SF ACCESSORY USE THAT DOES NOT CONTRIBUTE TO EXIT LOAD
158 SF
980 SF
SUM OF OCCUPANCIES IN CORRIDOR = 224 OCCUPANTS
368 SF ACCESSORY USE THAT DOES NOT CONTRIBUTE TO EXIT LOAD
103 SF
706 SF
120 SF
675 SF
706 SF
706 SF
OCCUPANT LOAD AT STAIR 22.4" REQUIRED WIDTH ACTUAL STAIR WIDTH = 69"
OCCUPANT LOAD AT STAIR 22.4" REQUIRED WIDTH ACTUAL STAIR WIDTH = 57"
EATD 78'+179' = 257'
78' CPET
EXIT STAIR
SECOND FLOOR PLUMBING FIXTURES
MEN'S W/C'S 4 LAVS 4
WOMEN'S W/C'S 3 LAVS 3
1 SHOWER
2 DF'S
2 SERVICE SINKS
11 OTHER SINKS

② SECOND FLOOR & MECHANICAL PLATFORM CODE PLAN- 1 0,727 GSF

SCALE: 1/16" = 1'-0"

NORTH


155 SF
1,937 SF
124 SF
1 HOUR RATED KITCHEN EXHAUST DUCTWORK. RE: MECH DETAIL 5/M5.00
ACCESSORY USES THAT DOES NOT CONTRIBUTE TO EXIT LOAD
EGRESS WIDTH = 68" ALLOWABLE OCCUPANCY THROUGH EXIT = 453 OCCUPANTS MAX
57 SF
A3 OCCUPANCY
EGRESS WIDTH = 68" ALLOWABLE OCCUPANCY THROUGH EXIT = 453 OCCUPANTS MAX
ACCESSORY USES THAT DOES NOT CONTRIBUTE TO EXIT LOAD
EGRESS WIDTH = 34" ALLOWABLE OCCUPANCY THROUGH EXIT = 226 OCCUPANTS MAX
10.7" REQUIRED EGRESS WIDTH ACTUAL OPENING WIDTH = 68"
81 SF ACCESSORY USE THAT DOES NOT CONTRIBUTE TO EXIT LOAD
145 SF
592 SF
201 SF
117 SF
✳129
161 SF
124 SF
18.45" REQUIRED EGRESS WIDTH ACTUAL OPENING WIDTH = 68"
A3 OCCUPANCY
1,392 SF
294 SF
270 SF
679 SF
445 SF
ACCESSORY USE THAT DOES NOT CONTRIBUTE TO EXIT LOAD
1,549 SF ADMIN SUITE
2,110 SF ADMIN SUITE
663 SF
634 SF
585 SF
589 SF
589 SF
FIRST FLOOR PLUMBING FIXTURES
MEN'S W/C'S 4 LAVS 4
WOMEN'S W/C'S 5 LAVS 4
4 DF'S
2 SERVICE SINKS
8 OTHER SINKS
37.65" REQUIRED EGRESS WIDTH ACTUAL OPENING WIDTH = 40"

CODE SYMBOL LEGEND

▪ ▪ ▪ ▪ ▪	PARTITION CONSTRUCTED TO RESIST PASSAGE OF SMOKE, RATING IS ALSO INDICATED ON REFLECTED CEILING PLANS.
— — —	ONE-HOUR RATED WALL. RATING IS ALSO INDICATED ON REFLECTED CEILING PLANS. NOTE: WALLS SHALL BE LABELED AS 1 HOUR RATED. REFER TO REFLECTED CEILING PLANS FOR MORE INFORMATION.
◁	EXIT LOCATION
⬡3	NUMBER OF OCCUPANTS
⑪②	NUMBER OF OCCUPANTS AT EGRESS COMPONENT
✳	THE LECTURE HALL EGRESS ANALYSIS IS BASED ON A MINIMUM 208 SEATING CAPACITY. SEATING CAPACITY WILL NOT EXCEED 299. ANALYSIS IS INTENDED TO DEMONSTRATE THAT THE EGRESS WIDTH PROVIDED EXCEEDS THE REQUIRED WIDTH FOR 208-299 OCCUPANTS.

① FIRST FLOOR CODE PLAN- 22,591 GSF

SCALE: 1/16" = 1'-0"

NORTH

OZ ARCHITECTURE

CODE SUMMARY

I. PROJECT INFORMATION

Project Name:	CMC-Summit	Project #:	56084
Location:	Breckenridge		
Authority Having Jurisdiction:			
Owner:	Colorado Mountain College		
Building Code:	International Building Code	Year:	2003
Mechanical Code:	International Mechanical Code (IBC)	Year:	2003
Plumbing Code:	International Plumbing Code (IBC)	Year:	2003
Electrical Code:	International Electric Code	Year:	2005
Energy Code:		Year:	
Local Amendments:		Year:	
Fire Code:	International Fire Code (IBC)	Year:	2003
Other:		Year:	
Code Analysis by:	Dan Miller	Date:	11/19/07

II. USE AND OCCUPANCY CLASSIFICATION

1. Occupancy Classification
302.1 A. Primary Occupancies: B- Educational occupancies above the 12th grade. A-3 (Assembly)
302.1.1 & Tbl 302.1.1 B. Incidental Use Areas: Storage rooms over 100 sf - Paritions to be constructed to resist smoke per section 302.1.1.1
302.2 C. Accessory Use Areas (<10%):
302.3.2 2. A. Nonseparated Uses: Yes, B occupancy and A-3 occupancy
B. Most Restrictive: A-3 Occupancy
Chapter 4 4. Special Detailed Requirements Based on Use and Occupancy:
Platforms (Lecture Hall) - 410.4 "Platform Construction "
Section 414.2 Control Areas. Sciene Lab and Prep room will be separated by a 1-hour fire barrier.

III. ALLOWABLE AREA & HEIGHT

1. Area of Proposed Building (Gross Floor Area)

First Floor	22,591	SF
Second Floor	10,727	SF
Mechanical Platform	541	SF
Fourth Floor		SF
Total	33,859	SF

2. Proposed Number of Stories: 2
Table 503 3. Proposed Construction Type: II-B
4. Allowable Height of Building
Table 503 A. In feet: 55.0
Table 503 B. In stories: 2
504.2 C. Increase allowed: Yes
D. Total allowable height/ stories: 25.0
5. Allowable Floor Area (Based on Occupancy & Construction Type)
507 A. Unlimited Area Allowed? No
Table 503 B. Allowable Area per Floor: 9,500 GSF
506.2 C. Increase for Frontage:

$I_f = 100\ [F/P - 0.25]\ W/30$ — I_f = 75 %

where:

I_f = Area increase due to frontage (percentage)
F = Building perimeter which fronts on a public way or open space having 20' open minimum. — 941 feet
P = Perimeter of entire building — 941 feet
W = Minimum width of public way or open space — 30 feet

506.3 D. Increase for Separation:
I_s = Area increase due to sprinkler protection — I_s = 200 %

506.1 E. Total Allowable Area per Floor:
$A_a = A_t + [A_t I_f / 100] + [A_t I_s / 100]$ — A_a = 35,625 SF/floor
where:
A_a = Allowable area per floor (SF)
A_t = Tabular area per floor — 9,500 SF
I_f = Area increase due to frontage (percentage) — 75 %
I_s = Area increase due to sprinkler protection — 200 %

F. Total Allowable Area per Building:
A_a = 35,625 SF/ Floor
× 2 Floors
Total 71,250 SF

705 6. Fire Walls Req'd to Separate Areas? No
508 7. Special Provisions? No

IV. TYPE OF CONSTRUCTION/FIRE-RESISTIVE REQUIREMENTS

1. Building Elements:
602 & Tbl 601 A. Structural Frame: 0
602 & Tbl 601 B. Bearing Walls - Exterior: 0 or greater of item 2. below
602 & Tbl 601 C. Bearing Walls - Interior: 0
602 & Tbl 601 D. Non-bearing Walls - Exterior: 0
602 E. Non-bearing Walls - Interior: 0
602 & Tbl 601 F. Floor Construction: 0
602 & Tbl 601 G. Roof Construction: 0
Table 602 2. Exterior Wall Based on Fire Separation Distance (but cannot be less than B. above):
A. <5' 1
B. 5' - 10' 1
C. 10' - 30' 0
D. >30' 0
704.5 3. Exterior Walls with Fire Separation Distance > 5' need only be Rated for Fire Exposure from Inside.
4. Construction Classification:
602.2 603.3 Type II - structural frame, walls, partitions, floors, and roofs required to be noncombustible. See 603.1 for areas where combustible materials are allowed.
5. Allowable Area of Openings (Based on Fire Separation Distance):
704.8.2 A. Unlimited in one story building?
Tbl 704.8 B. Based on Table 704.8:

Distance	Unprotected
0-3'	Not Allowed
3'-5'	15%
5'-10'	25%
10'-15'	45%
15'-20'	75%
20'-25'	No Limit
25'-30'	No Limit
>30'	No Limit

714.2 6. Exterior Wall Openings Shall be Protected According to Table Below:

Wall Rating	Opening Rating
3 & 2 hour	1-1/2 hour
1 hour	3/4 hour

714.2 C. Openings shall be protected according to table below.

Wall Rating	Opening Rating
3 & 4 hour	3 hour
2 & 1-1/2 hour	1-1/2 hour

706 9. Fire Barriers:
706.4 A. Shall extend from top of floor below to underside of deck above.
706.6 B. Openings shall be protected according to table below and limited in area.
706.4 C. Continuity must be maintained (N/A exceptions exist)
706.6 D. Exterior walls to match rating otherwise required for exterior wall, not the fire barrier
706.7 E. Openings 120 sf max. per opening (not limited if sprinklered), & 25%max. width of wall (not limited to either if ASTM tested and fire rating=wall rating) also per Section 714

Table 715.3

Wall Rating	Opening Rating
3 & 4 hour	3 hour
2 & 1-1/2 hour	1-1/2 hour
1 hour	3/4 hour

707 10. Shaft and Stair Enclosures:
707.4 A. Rating:
714.2 B. Opening Protection: 1 1/2-hour in 2-hour shafts, 1-hour in 1-hour shafts
708 11. Fire Partitions:
708.3 A. Rating: 1-hour (except corridors according to 1004.3.2.1 and some dwelling units & guest rooms)
708.4 B. Shall extend from top of floor below to underside of deck or floor/ ceiling assembly above (with exceptions)
708.4 C. Fire blocking required in combustible construction where partition does not extend to deck.
714.2 D. Opening Protection: 20-minute at 1-hour corridors, 3/4-hour at other fire partitions
709 12. Smoke Barriers:
709.3 A. Rating: 1-hour (except 0.1" steel at Group I-3)
709.4 B. Shall extend from top of floor below to underside of deck above (with exceptions)
715.3.3 C. Openings: 20-minute (except opposite swinging doors in Group I-2)
715.3 13. Fire Door Assemblies:
715.3.4 A. Only doors into exit enclosures require temperature end-point rating (exempt in sprinklered buildings)
715.3.4.1 B. If glazing is provided in door into exit enclosure, it must either be tested as a component of the door, or be less than 100 square inches (max. dimen. 10").
715.3.6 C. Glazing is prohibited in doors in fire walls, except doors serving as a horizontal exit may have 100 sq. in.
715.3.6.1 D. Glazing is limited to 100 sq. in. in doors having rating of 1-1/2 hours in fire barriers.
Table 715.4.3 E. If glazing is provided in any other fire door it must either be wire glass complying with limits in table below or comply with NFPA 80.

Door Rating	Wire Glass Limit
20 Minute	No Limit
3/4 Hour	1296 sq. in.
1 & 1-1/2 Hour (Int.)	100 sq. in.
1 & 1-1/2 Hour (Ext.)	None Allowed
3 Hour	None Allowed

715.4 14. Fire-protection-rated Glazing:
A. Glazing in fire barriers & fire partitions shall be wire glass in steel frame limited to 1296 sq. in. (54" max. dimen.) or comply with NFPA 80 & NFPA 257 and be 45-minute rated.
Table 715.4 706.7 B. Glazing in fire barriers & fire partitions is allowed in walls only up to 1-hour rated and is limited to 25% of the area of the wall.
715.2 15. Fire-resistance-rated Glazing: Allowed when tested as part of the wall assembly (rating equals wall rating).
716 16. Fire & Smoke Dampers:
716.5 A. Required locations:

	Location	Damper Type	Exceptions (No Damper Required)
716.5.1	Fire Walls	Fire Only	
716.5.2	Fire Barriers	Fire Only	Rating 1 hour or less, other than Group H & sprinklered
716.5.3	Exit Enclosures	Duct Not Permitted	
716.5.3.1	Shaft Enclosures	Fire & Smoke	Subducts, smoke control ducts, or parking garage shafts
716.5.4	Fire Partitions	Fire Only	Corridors and tenant separations in sprinklered buildings or Where duct < 100 sq. in., .0217" gage, has no opening to corridor, duct above ceiling, and doesn't terminate at wall
716.5.4.1	Corridors	Smoke Only	Buildings equipped with smoke-control systems or Where duct is 0.019" gage and has no opening to corridor
716.5.5	Smoke Barriers	Smoke Only	Single smoke compartment with steel duct

Table 716.3.1 B. Ratings: 3-hour in 3-hour or greater assemblies, 1-1/2-hour elsewhere
716.4 C. Access is required and shall be labeled by 1/2" letters reading SMOKE DAMPER or FIRE DAMPER.

V. FIRE PROTECTION SYSTEMS

903.2 1. Automatic Sprinkler System Required Based on Occupancy? Yes NFPA 13
903.2.12-15 2. Automatic Sprinkler System Required Based on Special Condition? Yes (903.2.12.2)
903.3.1.2 3. NFPA 13R System Allowed in Group R-1 & R-2 Buildings up to 4 Stories.
903.3.2 4. Quick-response Sprinklers Shall be Installed at: Group I-2 Sleeping Rooms, Group R- & I-2 Dwelling Units & Sleeping Rooms, and Light-haard Occupancies per NFPA 13.
905.3.1-6 5. Standpipe Class Required: No
906.1 6. Portable Fire Extinguishers as Required by the *International Fire Code*.
907 7. Fire Alarm and Detection Systems:
907.2.1-23 A. Automatic alarm required based on occupancy or special condition? No
907.2.1-23 B. Manual alarm required based on occupancy or special condition? Yes
907.2.1-23 C. Automatic detection req'd based on occupancy or special condition? No

VI EXIT REQUIREMENTS

Table 1004.1 Table 1005.1 1018.1 Occupant Load & Exits Required

Floor	S.F.	Use	Occupant Load Factor	Number of Occupants	Total Occupants per Floor	Number of Exits Required*	Total Exit Width Req'd (in.	Total Stair Width Req'd (in
1st	4,776	Educ.	20	239				
		Lecture Hall		239	239		35.9	
	1,392	Lobby	7	199	93		14.0	
	4,045	Business	100	40				
	1,937	Kitchen	200	10				
	1,430	Stor/ Mech	300	5				
			Total	732	[illegible]			[illegible]
[illegible]	4,169	Educ	20	208				
	927	Business	100	9				
	1,570	Stor/ Mech	300	6				
			Total	224		[illegible]	0.0	
Mech Plat	541	Mech	[illegible]	2	[illegible]		0.3	
Totals	20,787			91				

Accessible means of egress
A. Quantity required [illegible]
B. Buildings with 4 or more stories N/A
C. Accessible Enclosed stairway Not [illegible] #1
D. Accessi emergency operating and signaling devices required [illegible]

Doors
A. Minimum Size: [illegible]
B. Swing: in direction of exit travel when serving area with occ. load of 50
C. Floor Elevation same on both sides of door, excepting 1/2" floor finish thickness difference
D. Landings: 44" minimum
[illegible]

Stairways
A. Width: 44" minimum (stairs serving occupant load less than 50 may be 36")
B. Headroom 80" minimum
C. Riser [illegible]
D. Tread 11" minimum
E. Handrails required on each side ~~(except stairs less than 44")~~
34"-38" to top
[illegible]
E. Stairway to Roof [illegible]
F. Roof access: [illegible]

Exit Access
A. Separation:
B. Intervening Rooms allowed if room is accessory not high-hazard occupancy and discernable egress path provided
C. Common Path of Egress Travel: A Occupancy 75'
B Occupancy- 100' with sprinkler
1015. D. Travel Distance: A Occupancy- 250' with sprinkler
B Occupancy- 300' with sprinkler

1016 Corridors:
1016.2 A. Width: [illegible]
B. Construction
C. Fire resistance rating: 0
D. Dead end maximum: 50
E. Air movement Corridors shall not serve as supply, return, exhaust, relief, or ventilation air ducts or plenumbs
1016.5 F. Corridor continuity [illegible] (exceptions exist)

1017 Exits
A. Extent: enclosure to extend to exit discharge
1018 B. Number & Continuity:
1019 C. Vertical exit enclosures 2 hour rated if 4 or more stories
[illegible]

706 Fire Barriers
706.4 A. Continuity must be maintained (N/ A exceptions exist)
706.6 B. Exterior walls to match rating otherwise required for exterior wall, not the fire barrier
706.7 C. Openings 120 sf max. per opening (not limited if sprinklered), & 25%max. width of wall (not limited to either if ASTM tested and fire rating=wall rating)
[illegible]

Opening Protectives
A. Ratings [illegible]

VII ACCESSIBILITY

Ch. 11

VIII PLUMBING FIXTURE REQUIREMENTS

Tbl 2902.1 Fixtures Required:

	Occupancy		WC/ Male	WC/ Female	Lav/ Male	Lav/ Female	DF	Service Sink
A-3	239	Lecture Hall	1.0	1.8	0.6	0.6	0.5	
A-3	199	Lobby	0.8	1.5	0.5	0.5	0.4	1.0
B	40	Admin/ offices	0.4	0.4	0.5	0.5	0.4	
Educ	239	Classrooms	2.4	2.4	3.0	3.0	2.4	1
Total Lev 1			4.5	6.2	4.6	4.6	3.7	2
B	[illegible]	Admin/offices	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	1.0
Educ	[illegible]	Classrooms	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	
Total Lev 2			[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
Total			[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Fixtures [illegible]

	WC/Male**	WC/ Female**	Lav/Male**	Lav/Female**	DF	Service Sink
Level	4	5	4	4	4	2
Level	4	3	4	3	2	2
	8	8	8	7	6	4

OZ ARCHITECTURE
3012 Huron Street
Suite 100
Denver, Colorado 80202
phone 303.861.5704

COLORADO MOUNTAIN COLLEGE

107 DENISON PLACER ROAD
BRECKENRIDGE, CO

RECORD DOCUMENTS

These Record Documents have been prepared based on information (marked-up prints, drawings and other data) provided to the Architect by others. The Architect has not verified the accuracy and/or completeness of this information and shall not be responsible for the errors or omissions that may be incorporated as a result of erroneous information provided by others. These documents are for the Owner's convenience and cannot be used without written permission.

7/24/09

PROJ. NO. 56084.00
DRAWN: DM
CHECKED: DM

DATE: 7.24.09
REVISIONS:

©OZ ARCHITECTURE

RECORD DOCUMENTS

SHEET TITLE:
CODE PLANS

SCALE: AS NOTED
SHEET NUMBER:
A0.1